

JobPilot Technical Brief

BAX-423 Final Project Option B: Smart Job Matcher

1. System Overview

JobPilot is a runnable Streamlit application that recommends jobs from a 27,423-posting multi-country real snapshot extracted from the professor-recommended Techmap/Kaggle jobs dump. The system supports the six required capabilities: ingestion, resume/profile intake, dense retrieval, multi-stage ranking, adaptive feedback learning, top-job download, explanations, analytics, and tailored resume generation.

Architecture: Techmap/Kaggle zip stream parser -> normalized multi-country job snapshot -> simulated stream replay -> SQLite dedup store -> text normalization and skill extraction -> TF-IDF plus deterministic random-projection dense embeddings -> country-aware candidate filtering -> nearest candidate retrieval -> multi-stage ranker -> Streamlit UI with explain, feedback, analytics, CSV export, and resume draft.

Layer	Implementation	Production Upgrade
Data	27,423-row real Kaggle/Techmap sample across countries	Added field, seeded, deduplicated, and filtered with GPT/BigQuery.
Model	Dense vectors from TF-IDF + random projection; country filter	Vector DB, cosine similarity, chunking, retrieval, evaluation, classification, and location, and
Serving	Streamlit app with persona controls, recommendation cards	Cloud, RAG, feedback, and managed secrets.
Monitoring	Persona pass table and benchmark tab visible in app.	Logging, latency/error metrics, ranking-quality alerts.

2. BAX-423 Techniques and Benchmarks

Technique A: streaming-style ingestion and deduplication. The project streams the large Techmap/Kaggle zip without extracting the full 50GB JSON file, samples 27,423 real postings across multiple countries and fields, writes a normalized CSV, then replays postings in batches into SQLite and skips duplicate job_id values. This mirrors the lecture's prototype-to-production data layer: the app does not depend on live API calls during demo, but the boundary is ready for a real stream and current API refresh.

Technique B: embedding-based retrieval plus multi-stage ranking. Jobs and profiles are represented as TF-IDF plus random-projection dense vectors. The system first hard-filters to the persona's preferred countries, retrieves candidates, then adjusts for skill overlap, target-role alignment, salary/location preferences, user feedback, and dealbreaker penalties.

Persona	Keyword Skill Overlap	Multi-stage Skill Overlap	Keyword Violations	Multi-stage Violations
Aisha	1.0	0.4	5	0
Marcus	1.0	0.526	4	0
Priya	1.0	0.56	4	0
Kenji	1.0	0.525	0	0
Sofia	1.0	0.633	0	0
Lukas	1.0	0.504	3	0
Mei	1.0	0.282	3	0
Arjun	1.0	0.35	3	0
Nora	1.0	0.358	3	0
Elena	1.0	0.75	5	0
David	1.0	0.59	0	0
Leila	1.0	1.0	0	0
Maya	1.0	0.85	1	0
Omar	1.0	1.0	1	0

Interpretation: keyword matching can produce high overlap but ignores country and hard constraints. The multi-stage pipeline reduced top-10 hard dealbreaker violations to 0 across the required and extended personas while preserving role relevance.

3. Core Capabilities

Requirement	How JobPilot Implements It
Job ingestion & streaming	CSV snapshot is replayed into SQLite in batches; duplicate job_id inserts are rejected. The UI exposes the replay result.
Profile intake & skill matching	Sidebar accepts PDF/DOCX/TXT resume upload or pasted profile text, skills, target roles, preferred countries, salary, and dealbreaker.
Embedding retrieval	Profile and job text are embedded into dense vectors using TF-IDF plus random projection. Cosine similarity retrieves candidate jobs.
Multi-stage ranking	Preferred countries are applied before retrieval; final score combines embedding similarity, skill overlap, role alignment, salary/location.
Adaptive learning	Accept/reject/skip modifies skill-level preference weights and immediately changes ranking in the session.
Download top jobs	A CSV download button exports job_id, title, company, country, location, salary, apply link, and description for current top matches.
Explain & resume	Each recommendation shows score drivers and cautions; every job has a tailored resume draft and text download.
Batch analytics	Analytics tab shows top skills, salary by role family, and demand by country/location.

4. Persona Test Results

Persona	Countries	Result	Mean Score
Aisha	United States, Canada	PASS	0.716
Marcus	United States, Canada, United Kingdom	PASS	0.725
Priya	United States	PASS	0.671
Kenji	United States, Canada	PASS	0.588
Sofia	United Kingdom, Ireland	PASS	0.778
Lukas	Germany, Switzerland, Netherlands	PASS	0.583
Mei	Singapore, Hong Kong, Australia	PASS	0.584
Arjun	India, Singapore	PASS	0.561
Nora	Canada	PASS	0.536
Elena	United Kingdom, Ireland	PASS	0.781
David	Germany, Switzerland	PASS	0.544
Leila	United Arab Emirates, United Kingdom	PASS	0.658
Maya	India, United Kingdom, Vietnam	PASS	0.664
Omar	Canada, United States	PASS	0.855

All four required course personas plus ten additional personas pass in the current snapshot. The extended set covers analytics/ML, marketing, finance, healthcare operations, HR/recruiting, and sales across North America, Europe, and Asia-Pacific.

5. Validation and Acceptance Checks

The project was validated against the main grading surfaces: data volume, ingestion/deduplication, country-aware recommendation behavior, feedback response, resume intake, and multi-persona relevance. The checks below are designed to be reproducible from the submitted code and packaged data rather than relying on screenshots or manual claims.

Check	Expected Behavior	Observed Result
Dataset size	Use a large structured job dataset above the 500-record mark	27,423 normalized postings stored in <code>data/job_postings_sample.csv</code> .
SQLite replay	Stream postings into a structured store and deduplicate by title/job id	SQLite job id stores 27,423 rows; duplicate inserts are skipped by primary-key log.
Country hard filter	Recommendations should respect the user's preferred country	Ranking before scoring preferred-country jobs; impossible country lists return no rows.
Multi-stage ranking	Candidate generation, scoring, re-ranking, and hard penalties	Resolving between stages by skill, role, salary, location, feedback, and dealbreaker.
Feedback actions	Accept, skip, and reject should change the current session	Job posting feedback is stored and applied as positive, mild negative, or strong negative.
Resume upload	PDF, DOCX, and TXT profiles should be usable without copy/paste	Type parser parses uploaded files and fills the editable profile text area.
Persona coverage	Required personas should pass, with added personas beyond	Four MSB and many extended personas pass; added fields include marketing, financial, etc.

6. Data Quality Notes

The Techmap/Kaggle file is treated as a historical source of real postings, then reshaped into the fields needed by the application: title, company, country, location, description, skills, role family, salary range, remote flag, visa flag, and apply URL. Because the raw source does not consistently provide every downstream feature, several fields are inferred conservatively for ranking and demo purposes.

Field	Treatment in JobPilot
Country/location	Parsed and normalized so the app can apply preferred-country filters before retrieval.
Role family	Mapped from title and description keywords to support analytics and persona validation across multiple fields.
Skills	Extracted from posting text using a controlled skill vocabulary; used for overlap scoring and resume tailoring.
Salary	Kept when present and imputed from role family/country bands when missing; shown as an estimate, not an employer guarantee.
Remote/visa	Inferred from posting language when explicit metadata is absent; used as ranking signals and dealbreaker checks.
Description length	Long descriptions are capped for package size and app speed while retaining enough text for matching.

7. Suggested Demo Script

1. Start the Streamlit app, confirm the ingestion metric shows the SQLite replay and the 27,423-row dataset.
2. Select a persona from a non-US country preference, then verify all top recommendations stay within the selected country list.
3. Upload or paste a resume, adjust target skills/dealbreakers, and inspect the recommendation explanations.
4. Click Reject or Accept on one recommendation and rerun matching to show adaptive re-ranking.
5. Download top jobs as CSV, open one tailored resume draft, and briefly show the Analytics and Benchmarks tabs.

8. Limitations and Production Path

- The included data is a real September 2021 Techmap/Kaggle snapshot, not a live Adzuna/JSearch feed. A production version should refresh postings daily and store API provenance.
- Salary, company size, remote, and visa fields are imputed or inferred when the Kaggle posting lacks structured metadata; they are ranking aids, not verified employer facts.
- The resume generator is template-based and should be guarded by honesty checks, ATS review, and optional LLM prompt versioning before real use.
- Feedback learning is session-level. A deployed app should persist user-specific feedback, evaluate drift, and avoid overfitting to a few clicks.
- Current hosting is local/deploy-ready. To satisfy live hosting, push to GitHub and deploy ``code/app.py`` to Streamlit Cloud or Cloud Run.